GEK-83808

DRIVE SYSTEM INSTRUCTIONS

POWER SYSTEM STABILIZER
3S7932LA202G15

NOTE: INCLUDES CONTINUOUS WAVE FREQUENCY TRANSDUCER AND TORSIONAL FILTER.
FOR STANDARD 4-POLE GENERATOR

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

(GEK-83808)

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

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INTRODUCTION

The Power System Stabilizer (PSS) is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator speed or load, excitation becomes controlled as a composite function of voltage and frequency. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a three-phase signal from the generator potential transformers (PTs). The signal, which is frequency modulated, is passed through the continuous wave frequency transducer which then has an output frequency equal to the modulation frequency and an amplitude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristic, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer, signal conditioning, limiting and output circuitry. Figure 4 is a typical functional elementary diagram while Figures 5 through 11 are the printed circuit board elementary diagrams.

The Power System Stabilizer will apply the following transfer function to modify the output of the speed transducer for insertion into the voltage regulator:

$$G(s) = \frac{K\,(T_1S)\,(1 + T_2S)\,(1 + T_4S)}{(1 + T_1S)\,(1 + T_3S)\,(1 + T_5S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments can be made by calibrated dial settings which will be accurate to within 10% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the following typical ranges:

K = 1 to 100

T_1 = 0.1 to 50 seconds

T_2 = 0.2 to 1.5 seconds

T_3 = 0.02 to 0.1 seconds

T_4 = 0.2 to 1.5 seconds

T_5 = 0.02 to 0.1 seconds

RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt, the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to at least 5°C above ambient temperature to prevent condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

ac	Alternating Current
Amp	Amperes
Db	Decibels
dc	Direct Current
FET	Field Effect Transistor
Hz	Hertz
IC	Integrated Circuit
LED	Light Emitting Diode
Ma	Milliampere
Mv	Millivolt
PP	Peak-to-Peak
PT	Potential Transformer
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage
vco	Voltage-Controlled Oscillator
VDC	Direct Current Voltage

DESCRIPTION (COMPONENTS)

The Power System Stabilizer consists of:

3S7932LA202G15 Panel

The Power System Stabilizer is normally mounted in either the regulator cubicle or exciter cubicle and contains the following:

NUMBER	NAME	QUANTITY
44C331856-G01	Continuous Wave Frequency Transducer	1
44C331857-G01	Signal Conditioning Board	1
44C331859-G01	Washout and Output Board	1
44C331863-G02	Relay Board	1
44C331867-G01	Ceiling Sensing Board	1
44C372640-G02	Band-Reject Filter	1
44C331862-G01	Power Supply Board	1
3S7932MA321G9	Power Supply Panel(±36VDC)	1
3S7932MA321G10	Power Supply Panel (24VDC)	1
3S7932MA321G14	Transformer Panel	1

INSTALLATION

Location and Connections

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. (Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.)

Initial Checks

1. Wire check all external connections to the Power System Stabilizer. Be sure that wire size and type are adequate for the expected signal levels. (Refer to the connection diagram furnished with the excitation equipment.)

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* CAUTION *
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DO NOT HI-POT OR MEGGER ANY CIRCUIT, AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Insure that all printed circuit boards are fully inserted.

OPERATION

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* CAUTION *
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MISOPERATION OF THE POWER SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

Initial Operation

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* CAUTION *
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THE POWER SYSTEM STABILIZER MUST BE OUT OF SERVICE OR THE REGULATOR IN THE MANUAL MODE FOR ALL TESTS THROUGH PARAGRAPH 9 (EXCEPT PARAGRAPH 3 AS NOTED). FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR,

1. Input to Continuous wave Frequency transducer. With the turbine generator at rated speed, close switch L1SW to check input to Continuous Wave Frequency Transducer board (LOC 1101). This can be done by checking the voltages from common to test points TP2, TP3, and TP4 on that board. These are marked "A", "B", and "C", respectively. These voltages should be 2.1 ± 0.2 volts, RMS. They should be 120° apart, with "A" leading "B", and lagging "C". Phase sequence must be correct for proper operation.

   Open switch L1SW and pull the Continuous Wave Frequency Transducer board (LOC 1101).

2. Power Supply Tests. Turn on ±15 VDC power using the toggle switch mounted on the faceplate of the Power Supply board (LOC 1111). Check that the power supply voltages are correct. Adjustment of the ±15 volt supply may be made with board mounted rheostats (refer to Theory of Operation section). The current limit and the crow bar settings are factory set and need no further adjustment.

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* CAUTION *
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ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. INSURE THAT THE REGULATOR IS IN THE MANUAL MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

3. Relay Board Tests. If the generator is in the automatic mode of operation, transfer to manual. Pull the Relay board (LOC 1112). The Relay board is designed to allow the PSS to be used with many excitation systems. Board mounted shorting clips and rheostats are used to adapt the relay board to the particular excitation system. (Refer to the drawings furnished with the equipment for the settings of F1S, F2S, F3S, F4S, F1P and F2P.) Check settings and readjust if necessary. Insert Relay board. Pull Washout and Output board (LOC 1108). Put shorting clip C5S on terminals 1, 2 (not GRD). Insert Extender board in LOC 1108.

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* CAUTION *
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THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN MANUAL MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11, 12, to pins 3, 4 of the Extender board. This signal can be a dc signal or an oscillating signal (peak value should be less than 10 volts). With the PSS off (as evidenced by the "OFF" LED on the Relay board being lighted) no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" LED will not stay energized, press the reset pushbutton mounted on the faceplate of the Ceiling Sensing board. If the "ON" LED will still not stay energized, check that the permissive contact is closed (usually 52G line breaker - see sheet 4 of Figure 3). If the unit is off line, this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" LED should be lighted. Remove Extender board. Insert Washout and Output board in LOC 1108. On Ceiling Sensing board (LOC 1110), put the delay dial on zero, the positive limit dial on 500 and the negative limit dial on 500. Press the reset pushbutton on the Ceiling Sensing board. Adjust the gain for 1.0 by turning the dial marked "Gain K" on the Washout and Output board to zero. Connect a variable source of dc voltage to "IN 1" of the Washout and Output board and set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the Relay board. A voltage should exist at the test jack on the Relay board labeled "Out to Regulator". Increase the voltage at "IN1" of the Washout and Output board to ≈6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" LED to be lighted. Try to reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the on position. Remove the voltage from IN1. Press the reset pushbutton on the Ceiling Sensing board. Put the PSS in service (with the generator control in manual) by pressing the "ON" pushbutton. The PSS should latch in the on position and the "ON" LED should be lighted. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" LED should be lighted. Some installations may have customer controls for the "PSS" mounted in the control room. Check that any external customer controls can be used to insert and remove the PSS. Leave the PSS in the off position.

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* CAUTION *
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THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AUTOMATIC MODE. INSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. INSURE THAT THE CONTINUOUS WAVE FREQUENCY TRANSDUCER IS PULLED. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

The Relay board contains the necessary interface circuitry so that a one volt change in PSS output will cause a one percent change in generator terminal voltage. With the PSS off, a small step of voltage (≈ 1 VDC) should be inserted into IN1 of the Washout and Output board. The regulator should be in the automatic mode of operation with the generator preferably off-line. The test may be satisfactorily performed on-line if the reactive current compensator (if any) is bypassed or switched to zero and if the voltage regulator of any other generator units paralleled at terminal voltage are temporarily turned to manual control. Place the PSS in service. Measure the change in the generator terminal voltage. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt ±0.1. The minus sign (terminal voltage decreases for a positive input voltage) arises because the signal is inserted into IN1 which bypasses the washout stage. F1P (on the Relay board) can be used to adjust the gain, if necessary. Remove the voltage at IN1.

An attempt has been made to interface with the voltage regulator in such a way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the Relay board has been furnished for this purpose. With the voltage regulator in manual (or test if a test position is available on the control switch) pull the Continuous Wave Frequency Transducer (LOC 1101). Put the PSS in service using the pushbuttons on the Relay board to minimize the change in ac regulator output voltage when the PSS is inserted.

Remove the PSS from service.

4. Continuous wave Frequency Transducer Tests. Check the operation of the Continuous Wave Frequency Transducer board (LOC 1101). Make sure switch L1SW is open, and connect a 3-phase, 115VRMS, 60 Hz signal to points L25A, L26A, and L27A. Phase sequence must be in that order. Insert the Continuous Wave Frequency Transducer board. The output, measured at TP5, should be zero. This test point is mounted on the faceplate of the C.W.F transducer board and is marked OUT. If necessary, reduce this voltage to zero using potentiometer R2, marked NULL on the C.W.F. transducer board. Remove the 3-phase, 115VAC source.

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* CAUTION *
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THE PSS MUST BE OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

5. Frequency Response Tests. On the Signal Conditioning board (LOC 1102), the Washout and Output board (LOC 11081, and the Band Reject Filter (LOC 1109) record all pot settings and jumper positions prior to testing so that controls can be reset to former conditions if desired.

Set the Signal Conditioning board (LOC 1102) for lead time constants Of 0.26 seconds and lag time constants of 0.03 seconds. This can be accomplished by the following adjustments:

TCT2V = 0 turns

TCT2F = 0.9 turns

TCT3 = 1.5 turns

TCT4 = 0.9 turns

TCT5 = 1.5 turns

BIS jumper should be on l-2 (GRD).

On the Washout and Output board (LOC 1108) set the washout time constant for 2 seconds and the gain for 20. This can be accomplished by placing the TCT1 dial on 6.8 and the gain control at 3.8.

Jumpers should be as follows:

C1S	2-3	
C2S	2-3	(150K)
C3S	l-2	(1-50)
C4S	l-2	(1-50)
C5S	2-3	(GRD)

Check the Band-Reject Filter (LOC 1109) to confirm that the shorting clips on 1TB, 2TB, 3TB and 4TB are on terminals 1, 2. The shorting clip on 5TB should be on terminals 2, 3. Pull the Continuous Wave Frequency Transducer board (LOC 1101). Insert the Extender board in place of the Transducer board. Connect an audio oscillator to terminal 11, 12 and ground terminal 3 or 4 of the Extender board. Measure the frequency response of the PSS main channel from 0.16 Hz to 143 Hz measuring oscillator output and the output of the Washout and Output board. The latter can be measured on the "Out 1" test jack of the Washout and Output board. The oscillator output level should be held to those shown in Figure 1A to avoid saturation of the amplifiers record data and compare gain to values shown in Figure 1A.

6. Ceiling Sensing Tests. Check out the Ceiling Sensing board (LOC 1110). Pull the Ceiling Sensing board and verify that the shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit (E2P) and negative limit (E3P) for 500 (5.0 turns). Set the time delay (E1P) on the Ceiling Sensing board for minimum (0). Ascertain that the PSS is off. Insert a variable dc voltage into IN1 of the Washout and Output board. Place the regulator in the manual mode of operation.

FREQ HZ	V(IN) VOLTS (ZERO TO PEAK)	V(OUT) VOLTS (ZERO TO PEAK)	GAIN	NOMINAL GAIN (1)
0.1	0.2			14.1
0.2	0.2			18.2
0.4	0.2			24.8
1.0	0.2			58.2
3.0	0.02			164.1
5.0	0.02			197.2
7.0	0.2			12.3
10	0.2			33.9
12	0.02			69.2
15	0.02			36.7
17	0.02			46.8
20	0.02			51.9
25	0.02			82.2
30	0.02			95.5
35	0.02			83.2
40	0.02			55.0
45	0.2			32.4
50	0.2			14.6
55	0.2			5.2
65	0.2			3.6
70	0.2			4.3
80	0.2			5.8
90	0.2			7.4
100	0.2			9.2

T_1 = 2 SEC. T2 = T4 = 0.26 SEC
T3 = T5 = 0.03 SEC.
GAIN = 20

Gain is measured from the output of the Continuous Wave Frequency Transducer to the output of the Washout and Output board. Use an oscilloscope to ascertain that nothing is saturating.

Figure 1A
Frequency Response of Main Channel

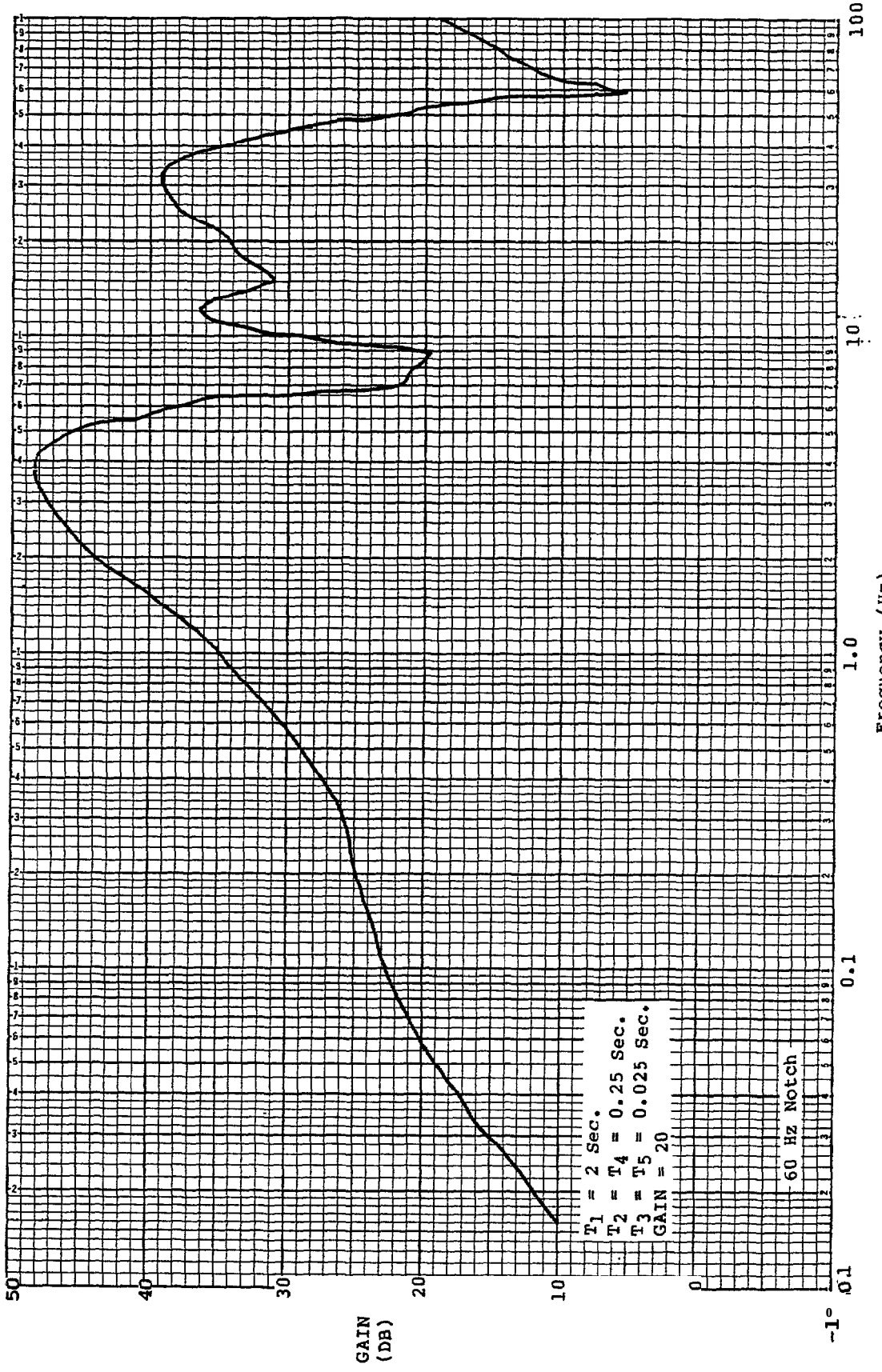


Figure 1B PSS
Gain vs Frequency
Excluding C.W.F. Transducer
Band Reject Filter for Standard 4-Pole Generator

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* *
* CAUTION *
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THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO DO THIS COULD RESULT IN DAMAGE TO THE GENERATOR.

Place the PSS in service by pressing the "ON" pushbutton on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" pushbutton being lighted. The voltage at Out 1 on the Washout and Output board should be 5 volts ±0.25 volts. Remove the voltage. Pressing the "ON" pushbutton on the Relay board should have no effect. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service. Reverse the voltage into IN1 of the Washout and Output board. The same procedure can be followed to verify that this part of the circuit is working. Adjust the time delay (TDP) on the Ceiling Sensing board for 300 (3 turns). Remove the voltage at IN1 of the Washout and Output board and press the reset pushbutton on the Ceiling Sensing board. Place the PSS in service (regulator in manual). Put a voltage into IN1 of sufficient magnitude as to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Remove the test voltage and reset the Ceiling Sensing board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

7. Washout and Output Tests. On Washout and Output board, set negative limit dial and positive limit dial on 500 (5 turns). Insert a variable dc voltage into IN1. Monitor the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the dc voltage slowly and note at what voltage the output fails to follow the input voltage (output voltage is limited). This voltage should be 4.75 volts ±1.5 volts. Reverse the voltage and perform the same test. The output voltage should be of the opposite polarity but same magnitude (within ±0.5 volts). Remove the voltage source and Extender board. Reinsert the Continuous Wave Frequency Transducer. Insure that the PSS is off. Place the regulator in the automatic mode of operation.

8. Maximum Gain Noise Tests. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

The adjustment range provided by the PSS can result in extremely high gain over a range of frequencies. For instance, with K set at 100, T2 and T4 set at maximum and T3 and T5 set at minimum, the maximum gain will be 120 db (1,000,000:1). Although these settings will probably never be used, a good check of PSS noise level can be obtained using them.

Make sure the PSS is out of service. On the Signal Conditioning board (LOC 11021, set TCT2F and TCT4 for 1000 (10 turns) and TCT3 and TCT5 for zero.

Pull the Washout and Output board (LOC 1108). Put the shorting clip on C5S on terminals 2, 3 (GRD). Put the shorting clip on C3S on terminals 2, 3. Insert Washout and Output board. Turn gain adjustment to 1000 (10 turns). The Continuous Wave Frequency Transducer board (LOC 1101) and insert the Extender board in its place. Short from pin 15 to pin 3 of the Extender board. Using an oscilloscope with a very slow sweep speed (1 sec/cm), determine the peak-to-peak output voltage (measured at OUT 1 of the Washout and Output board. This voltage should be observed for some time, perhaps up to a minute. This voltage should be less than 3 volts peak-to-peak. Remove the Extender board. Insert the Continuous Wave Frequency Transducer board in LOC 1101. Remove Washout and Output board. Place C5S shorting clip on terminals 1, 2 (not GRD). Place C3S shorting clip on terminals 1, 2. Return gain adjustment dial to zero (unity gain). Insert Washout and Output board. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

Initial Settings

The PSS has been designed to interface with various excitation systems. In order to permit compensation for widely different systems, adjustments are made using calibrated dials. On the Washout and Output board, set the gain dial to zero and C3S on terminals 1 and 2. Insure that the Relay board has the correct rheostat and switch settings, and that BIS on the Signal Conditioning board has the shorting link on terminals 1, 2 (GND).

On-Line Tests

Power System Stabilizers are employed to provide an additional input to the regulator to improve power system dynamic performance. They have been employed to solve two separate but related problems: local machine-system oscillations in the l-2 Hz range and inter-area oscillations $\leq$ 0.5 Hz. The fundamental function of the signal conditioning network of the PSS is to compensate for the phase lags of the system being controlled. For best contribution to system damping, the gain of the PSS should be maximized under the most sensitive conditions (usually at or near full-load). The desired stabilizing signal has as its reference the base system frequency which is slowly changing all the time. The washout function minimizes the effect of overall system speed changes on machine terminal voltage.

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* CAUTION *
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THE PSS HAS A WIDE RANGE OF ADJUSTMENT OF LEAD AND LAG TIME CONSTANTS. DO NOT ADJUST FOR A DIFFERENCE IN LEAD-LAG SETTINGS OF MORE THAN TEN TO ONE. LARGER DIFFERENCES IN SETTINGS CAN CAUSE EXCESSIVE AMPLIFICATION OF TORSIONAL SIGNALS AND RESULT IN DAMAGE TO THE TURBINE-GENERATOR.

The PSS has adjustable limits on the output signal so that the PSS will not completely override voltage regulator action. The output voltage limits are independently adjustable but should be set identically in order to prevent nonsymmetrical limiting which might bias the voltage regulator. Since the ceiling sensing function occurs after the adjustable output limit stage, the output voltage limits should be set at a higher voltage than the ceiling sensing limits (Refer to Figure 2).

In order to facilitate on-line frequency response testing of the excitation system, IN1 is provided on the Washout and Output board. With the shorting clip on C5S on terminals 1, 2, a drive signal can be inserted into IN1 which will produce a calibrated terminal voltage change through the regulator. For normal operation, the shorting clip on C5S should be on terminals 2, 3 (GRD) and the shorting clip on C1S should be on terminals 2, 3.

Figure 2 is provided so that dial settings can be chosen to realize the desired time constants and limit settings.

PRINCIPLES OF OPERATION

General

The Power System Stabilizer (PSS) is used to provide an additional input to the voltage regulator which, under certain conditions, can improve power system dynamic stability. A signal is obtained from the secondary of a 3-phase potential transformer whose primary is connected to generator terminal voltage. This signal is frequency-modulated, with the 60 Hz carrier being modulated by the frequency changes of the terminal voltage.

This signal is passed through a Continuous Wave Frequency Transducer to produce an output signal whose magnitude is proportional to the magnitude of the deviation in speed and whose frequency is equal to the speed deviation frequency. This demodulated signal is then passed through two stages of lead-lag compensations to provide phase shift compensation for the system, a twin-tee notch filter tuned to 60 Hz (30 Hz for an 1800 RPM machine), a washout stage to remove drift and slow system frequency changes and an adjustable gain stage to maximize the effect of the PSS. Limits are provided on the output stage so that during large system disturbances, the PSS will not interfere with voltage regulator action. A relay is provided for placing the PSS in service (and also removal from service). Interfacing is provided to properly scale the PSS output signal so that 1 volt from the output stage produces a 1% change in terminal voltage. A protective circuit is provided to detect a failure which causes the output to go to ceiling (Ceiling Sensing, board). Ten-turn rheostats with calibrated dials are furnished for all customer adjustments. Figure 4 contains an overall block diagram and a functional elementary of the PSS. Figures 5 through 11 are the elementary diagrams of the printed circuit boards.

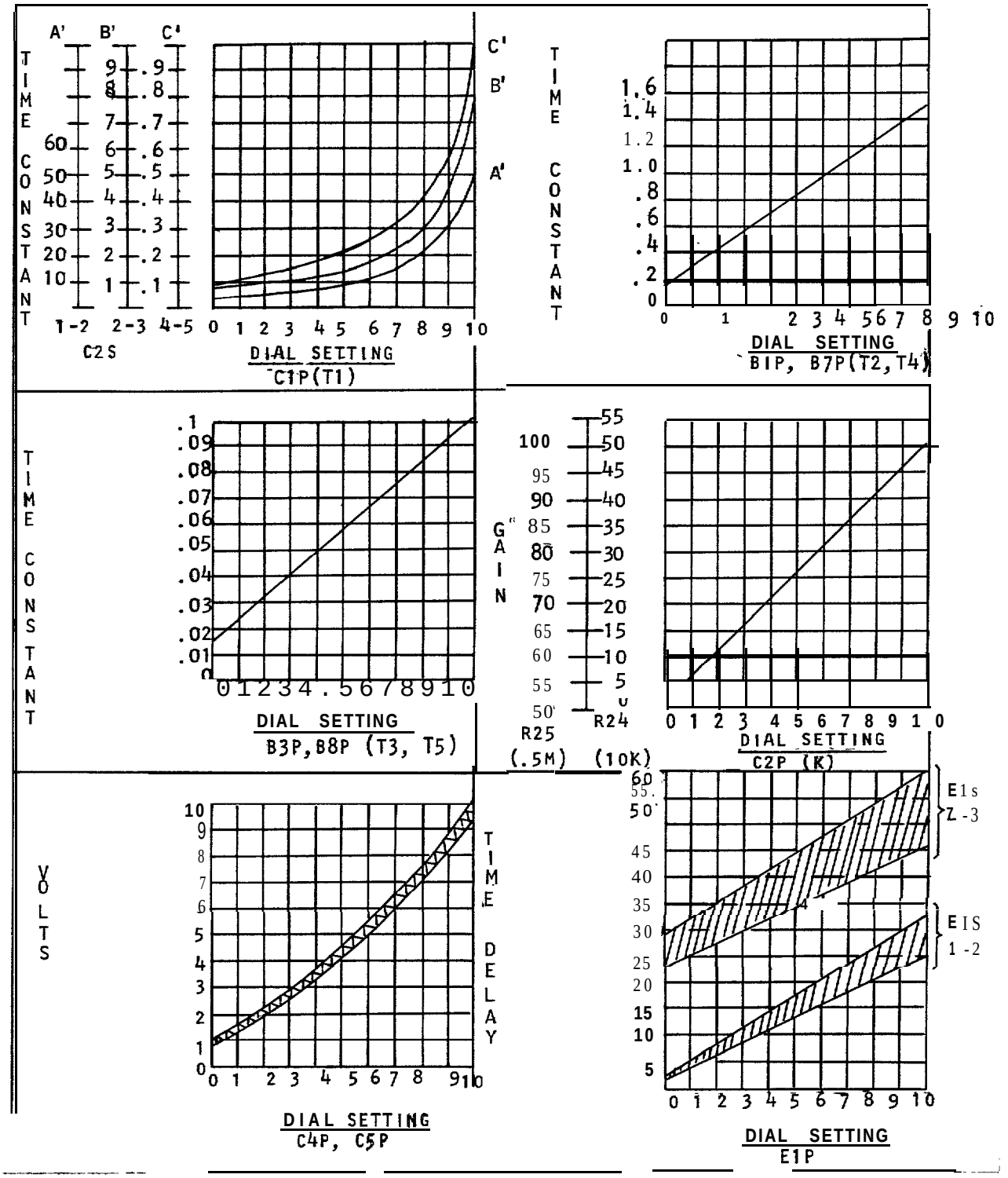


Figure 2
Parameter Value vs Dial Setting

Power Supply System (Figure 4, Sheets 4 and 5, and Figure 5).

The panel-mounted PSS includes both the ± 15 VDC and 24 VDC power supplies, The 115 VAC, 60 Hz power should be connected to L21 and L22. S1T and S2T transform the input voltage and provide isolation. The voltage is rectified by a single phase full-wave bridge, filtered to remove the AC component, limited by Zener diode action and then fed to series pass regulators. Fuses R1F and R2F, SCRs R1CR and R2CR, along with associated sensing and firing circuits form a "crowbar" circuit which will protect sensitive semiconductor components from overvoltage by blowing a fuse and thus removing the excess voltage. IC voltage regulators (G1A, G2A) are used to control the output voltage. The G2P and G4P mounted on the printed circuit board provide adjustment of the output voltage. The G1R, G1P, G2R, G3R provide adjustment of the output voltage. The G1R, G1P, G2R, G3R provide a current limit feature on the t15 volt regulator (a similar circuit is found on the -15 volt regulator). The current limit is of the foldback type, i.e., the maximum output current is reduced as the load on the regulator is increased above the full-load value. This insures that critical components in the regulator (series pass transistors) will not be damaged by a short circuit. The current limit can be adjusted using G1P and G5P. The positive limit is factory set at 1 amp and the negative limit at 750 Ma (these are not the short circuit current values). Integrated circuits J1A and J2A provide voltage sensing for the "crowbar" circuit. The J1P and J2P are factory adjusted to cause the "crowbar" circuit to operate at a supply voltage of 17 volts. The S1T and S2T have taps which will permit variations in the input voltage from the nominal 115 VAC value. (Refer to the diagram for the correct tap for a given input voltage.) In order to reduce the noise level to as low a value as possible, the ± 15 volt power supply is very tightly regulated and heavily filtered. The regulation is 0.1% and the ripple on the power supply should be ≤ 5 Mv P-P. The 24 volt supply does not require tight regulation since it is used to drive relays and energize lights. The S3T is used to isolate the 24 volt supply from the ±15 volt supply and to transform the input voltage to a lower voltage. This voltage is rectified by a full-wave bridge and then filtered. Regulation is provided by a Zener diode (S1Z). The 24 volt power supply is ungrounded while the ±15 volt supplies are grounded through a parallel resistor capacitor combination. The high impedance ground on the ±15 volt supplies allows the PSS to be used with both grounded and ungrounded systems without ground loop problems.

Continuous Wave Frequency Transducer Board (Figure 6)

The Continuous Wave Frequency Transducer is a three-phase, phase-lock-loop which tracks the input line frequencies, effectively providing frequency demodulation. The three line voltage signals reduced by PTs, are individually multiplied by the cutout of a voltage-controlled oscillator (VCO) to produce sum and difference phase error signals. Before multiplication, the output of the VCO is phase shifted to attain the proper phase relationship with each of the three phases. The three phase-error signals are summed and then passed through a low pass filter (an integrator) to remove the phase-sum terms in the multiplier output such that only the phase-difference terms remain. The frequency tracking is due to these phase difference terms. The signal is next passed through a lead stage to make the loop stable: the output of this stage is the frequency transducer output. To complete the loop, the output is fed back to the input of the VCO along with a summed-in DC bias used to null the VCO at 60 Hz. The transducer is adjusted to give a 1 volt change for a 1% change in input frequency.

The three phase voltage signals are inputs to the X multipliers of U1, U2, and u3. The Y multipliers come from the phase shifted VCO outputs, except for phase A which, not needing any phase shift, comes directly from the VCO. The 120° lag and lead needed for phases B and C are provided by the two amplifiers in U12. The VCO output (IA) along with its quadrature component (QA) and its merge (both available from the VCO circuitry) are scaled and summed by U2 such that the necessary phase shifts result. Potentiometers R3 and R5 vary the phase angles of the VCO output for phase B and C, respectively, and potentiometers R4 and R6 vary the amplitudes of the same respective signals. The outputs of the three multipliers are next summed at pin 7 of U14. The sum is next passed through a 120 Hz notch filter comprised of amplifiers U10 and U11. These amplifiers form a Biquad filter which has the following transfer function:

$$\frac{\frac{s^2}{\omega_o^2} + \alpha_N s + 1}{\frac{s^2}{\omega_o^2} + \alpha_D s + 1}$$

An unwanted 120 Hz signal caused by the voltage and phase unbalance is limited by the filter to 0.2 volt peak-to-peak. After being inverted by one amplifier in U9, the signal passes through the other U9 amplifier which provides a transfer function equal to K(1 + TIS)/S. This provides the low pass filter (integrator) necessary for frequency tracking and the lead stage necessary for loop stability. Jumper J2 provides a wide or narrow bandwidth setting, the former having an open loop crossover frequency in the 30 Hz to 45 Hz range with more than 45° phase margin, and the latter having an open loop crossover frequency of 15 Hz with the same phase margin.

The signal is now summed with a DC signal at the input to the VCO (pin 2 of U7) for nulling.

The VCO is implemented by simulating a resonant RLC circuit with the current and voltage signals tied together through pins 2 and 6 of U8. The Zener diode in the feedback of amplifier U8 causes the output to be a square wave. The X multipliers of U4 and U5, which are tied together and are the input to the VCO, alter the gain and consequently, the resonant frequency of the loop. The vco output, its inverse, and its quadrature component are tapped off pins 12 of U13, 10 of U13, and 12 of U14, respectively, and used as previously mentioned. All jumpers except J2 are for factory test purposes.

Signal Conditioning Board (Figure 7)

The Signal Conditioning board consists of two lead-lag stages and a twin-tee notch filter. In addition to the signal from the Continuous Wave Frequency Transducer, an extra input with separate lead time constant adjustment is provided for a voltage input (IN2). B1P provides adjustment of the first stage lead time constant, which is proportional to B1C, B1R and the setting of B1P. B3P provides adjustment of the lag time constant which is proportional to B9R, B4C and the setting of B3P. The dc gain of the stage is B9R/B1R, which is 1.0. B3R, B3C, and the equivalent resistance of B3P cause extraneous lead and lag time constants which have been minimized but may be significant when performing calculations at torsional frequencies. The second lead-lag stage is identical to the first except for the addition of an output voltage limit circuit formed by B1Z, B2Z, B5D, B6D and a drive circuit formed by B1Q, B21R, B22R, B3D and B4D to increase the output current capability of B2A. A twin-tee notch filter is placed between the first and second lead-lag stages to remove 60 Hz. The twin-tee is factory adjusted to produce greater than 50 Db attenuation at the notch frequency. The notch frequency is given by:

$$F = \frac{1}{2\pi(B11R + B4P)(B16C)}$$

Band-Reject (Biquad) Filter (Figure 8)

The Band-Reject (Biquad) Filter is a five-stage filter used to attenuate signals at the torsional resonant frequencies of the turbine generator. It provides unity gain outside the rejection band and approximately 40 Db attenuation in the rejection band. Figure 3 is a calculated attenuation versus frequency curve. In each stage, amplifiers 11A, 12A and 13A with associated components comprise a resonant circuit. The output of 12A is summed with the input signal in 14A to obtain an output signal. Signals not within the pass band of the resonant circuit are passed with no attenuation. Signals that fall within the pass band of the resonant circuit are passed and then subtracted from the original signal to give a reject characteristic. The transfer function for each stage is described by the following equation:

$$\frac{e_o}{e_i} = \frac{1 + \frac{\alpha_N}{\omega_o} S + \left(\frac{S}{\omega_o}\right)^2}{1 + \frac{\alpha_D}{\omega_o} S + \left(\frac{S}{\omega_o}\right)^2}$$

Where α_N, OLD, and ω_o are determined by the value of the resistors and capacitors used.

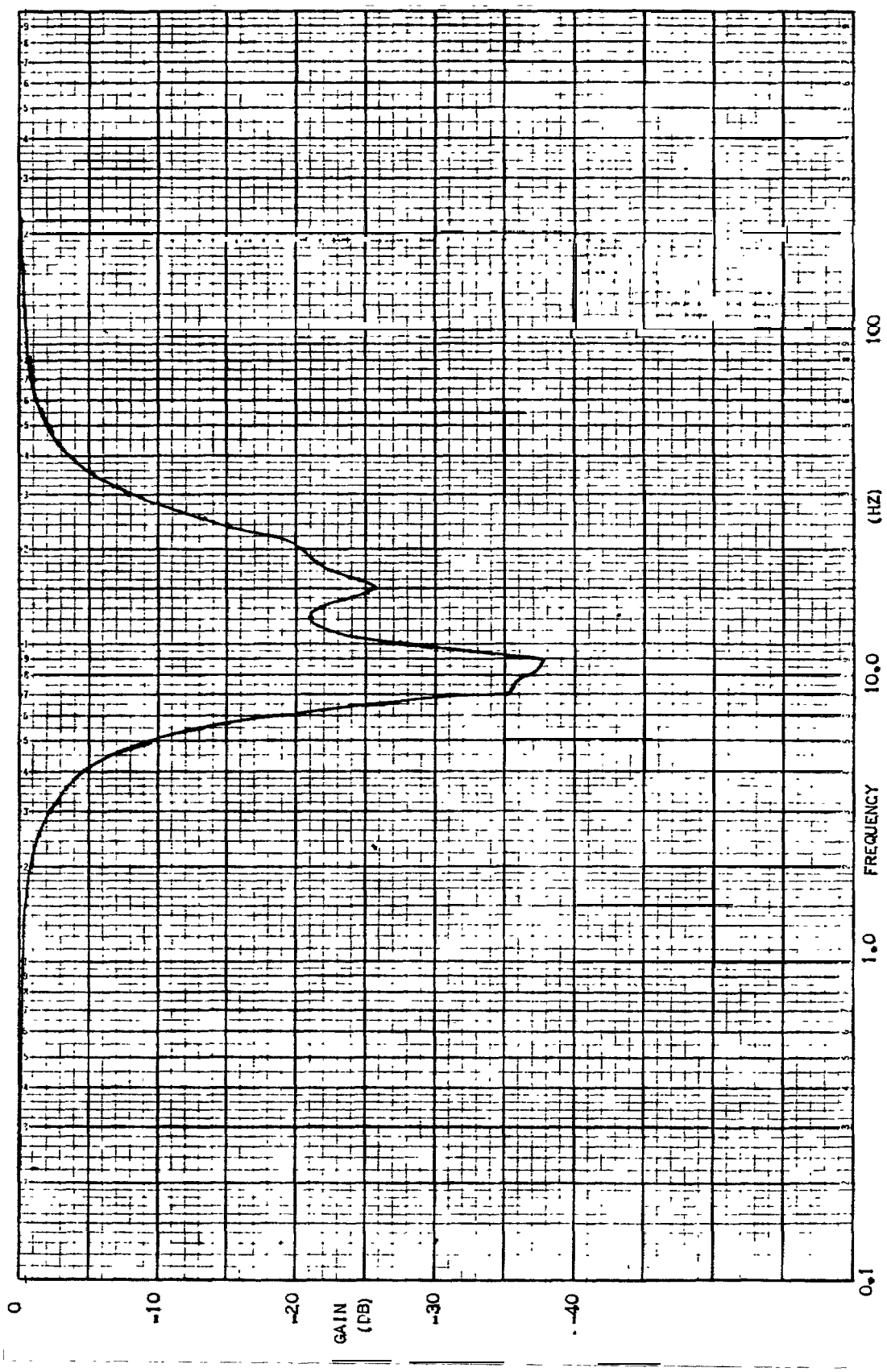


Figure 3
Gain vs Frequency
Band Reject Filter for Standard (4-Pole Generator)

Washout and Output Board (Figure 9)

The Washout and Output board contains a washout function, a gain function and a power amplification function. Operational amplifier C1A is a chopper stabilized device for low offset. C1A is basically a derivative circuit formed by C1C and the feedback resistor (C2R, C3R or C4R). C2C is added to make the derivative circuit only effective for low frequencies and C1P is used to provide adjustment of the effective feedback resistance and thus control the time constant. Coarse adjustment of the time constant is available through C2S. The C9D, C10D, Z1C, Z2C, C31R and C32R form a clamp to limit the output voltage of the washout stage. An extra input is available at IN1 which bypasses the washout stage. This allows insertion of a signal into the voltage regulator for calibration of the PSS or for frequency response testing of the system. The C5S is used to ground IN1 during normal use. Changing C5S shorting clip to terminals 1, 2 allows IN1 to be used for driving the regulator. The C2A is used as an adjustable gain stage with the gain being determined by C3S and C2P. The C3 is used to select a range of gain from 1 to 50 or 50 to 100. An adjustable limit is provided on the output of C2A by C1Q, C2Q, C15R, C16R, C1D, C2D, C4P, C5P, C13R and C14R. The output of C2A is used as the input signal for Ceiling Sensing board through pins 13, 14. The C3Q and C4Q are junction FETs used as a static switch. In normal operation, C1S should have the shorting clip between terminals 2, 3 (OUT). The C3Q will be conducting and C4Q will be non-conducting. The C3A is connected to a push-pull transistor stage (C7Q, C8Q, C9Q, C10Q) in order to increase the output current capability to ±100 Ma. The C3A and the push-pull stage are connected as a unity gain voltage follower.

Relay Board (Figure 10)

The Relay board is used to control insertion of the PSS into the regulator, provide the proper interfacing with different voltage regulators and provide interfacing with customer equipment. Meter M1 is mounted on the faceplate and can be used to insure that the PSS output is zero before insertion into the voltage regulator. "OFF" and "ON" pushbuttons are mounted on the faceplate and can be used to insert or remove the PSS from service. LEDs mounted on the faceplate indicate the status of the PSS. A line breaker contact (52G) should be used as a permissive contact so that the PSS is inoperative off-line. The pushbuttons have momentary contacts and so F93 is sealed in by an FAR and F93 contact. The F93A is an auxiliary relay used to increase the number of contacts of F93. FAR is controlled by the Ceiling Sensing board.

The F1R, F2R, and F1P are used to properly scale the PSS signal for insertion into the regulator. The F2P is used to dummy load the regulator so that a bump in excitation will not occur whenever the PSS is placed in service. (Refer to the diagram for the proper settings of F1P and F2P and switches F1S, F2S, F3S and F4S.)

Ceiling Sensinq Board (Figure 11)

The Ceiling Sensing board is used to detect a failure in the PSS which causes the output to be greater than a specified value for an adjustable length of time. Once the condition exists, the PSS cannot be restored to service until a reset pushbutton is depressed on the Ceiling Sensing board.

The E1A and E2A operate as comparators with E2P and E3P used to adjust the level at which a failure is detected. If the input voltage is less than the positive limit (E2P) and more positive than the negative limit (E3P) the voltage at E6TP is at a minimum. The voltage at E5TP is passed through an inverter in E1L and then a nand operation is performed on the signal from E6TP and the inverted signal from E5TP. A nand gate has its output in the high state unless all inputs are in the high state.

In this case, both inputs are in the high state so its output is in the low state. The output of this nand gate is passed through two inverters (no change in state) and is also used to trigger a one-shot (E3L). With a ceiling condition not detected, the one-shot trigger input will stay in the low state and the output (the complemented output is used in this case) stays in the high state. The inputs to the first nand gate of E2L are a low and a high, which means the output will be in the high state. Two of the nand gates are used in a cross-coupled fashion to allow latching whenever a ceiling condition is detected, the inverter output of E2L will be in the low state, turning on transistors E2Q and E3Q and energizing relay FAR on the Relay board.

If a ceiling condition exists (whether positive or negative), the second nand gate of E1L will have its input states changed to one low and one high. The output of this nand gate will go to the high state which triggers the one-shot (E3L) and changes the input on the first nand gate of E2L to a high state. Nothing happens however, because the output of the one-shot (E3L) is in the low state and remains there for a period of time determined by E7C, E1P and E8R or E9R (depending on the setting of E1S). After the timing interval, the output of the one-shot (E3L) goes to the high state and the output of the first nand gate of E2L goes to the low state. The output of the cross-coupled gates of E2L goes to the low state and will remain low even if the ceiling condition is cleared. The inverter output of E2L changes to the high state, which turns off E2Q and E3Q and drops out relay FAR on the Relay board, giving an alarm. To clear the alarm and place the PSS back in service, it is necessary to press the reset pushbutton (E1PB). This changes the input to one of the cross-coupled gates from a high to a low state which forces the output of the inverter to be in a low state. The cross-coupled gates latch in this state, the relay FAR is picked up and the alarm is cleared.

MAINTENANCE

The PSS is static, having no moving parts except for relays and instruments. Therefore, little maintenance should be required. If excessive vibration is present, all screw connections should be checked periodically. The Continuous Wave Frequency Transducer null adjustment should be checked periodically and readjusted if necessary.

TROUBLESHOOTING

1. Carefully check all wiring if problems are experienced at start-up.
2. If no response is obtained while making the frequency response test:
   a. Check that the power switch on the Power Supply board is on (if a Power Supply board is furnished).
   b. Check fuses S1F and S2F on the Power Supply panel.
   c. Check that all boards are fully inserted.
   d. Check for proper jumper connections on Washout and Output board.
   e. Check each board individually for a response.
3. If an incorrect signal is obtained in terminal voltage, check the switch and rheostat settings on the Relay board.
4. If the PSS will not stay in service:
   a. Reset Ceiling Sensing
   b. Check that permissive contact is closed.
5. Continuous Wave Frequency Transducer output stays at ceiling:
   a. Insure that generator speed is at rated.
   b. Readjust null.
6. An instability is observed with gain dial at zero:
   a. Measure gain to ensure that it is unity.
   b. Insure that an increase in frequency causes an increase in generator field voltage.
7. If a board is found to be defective, replace it and return the defective board for repair and return.

RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with complete description including use and location. Board components should not be replaced: order the entire board.

2. Complete nameplate data appearing in the assembly that included the component, including the serial number of the assembly.

3. If possible, data on original order on which equipment was first supplied, including all numerical references.

	TYPE	G01 B.R.	G02 B.P.	G03 B.R.	G04 B.P.	G05 B.R.	G06 B.P.
	11R	43.61	43.6K	94.71	94.7	55.04K	55.04K
	12R	43.61	43.6K	94.71	94.7	55.04K	55.04K
	15R	83.31	8.33K	100.01	10.0	125.0K	lb.6716
	16R	8.331	83.3K	10.01	400.0	16.67K	125.0K
	21R	35.01	35.0K	82.81	82.8	51.41K	51.41 K
	22R	35.01	35.0K	82.81	82.8	51.41K	51.41K
	25R	83.31	8.33K	50.01	16.7	125.0K	12.50K
	2bR	8.331	83.316	16.71	50.0	12.50K	125.0K
	31R	28.11	28.1K	73.61	73.6	48.05K	48.05K
	32R	28.11	28.1K	73.6K	73.6	48.05K	48.05K
	35R	83.31	8.33K	100.01	10.0	125.0K	16.67K
	3bR	8.331	83.3K	10.0K	100.0	16.67K	125.0K
	41R	22.51	22.5K	44.211	44.2	21.59K	21.59K
*	42R	22.51	22.5K	44.21	44.2	31 .09K	37 .09K
	45R	83.31	8.33K	50.01	a.33	100.0K	8.333K
	4bR	8.331	83.3K	8.331	50.0	8.333K	100.0K
	51R	18.01	18.0K	31.61	31.6	19.18K	19.18K
**	52R	16.01	18.0K	31.61	31.6	27.62K	27.62K
	55R	83.31	8.33K	33.31	a.33	100.0K	8.333K
	5bR	6.331	83.3K	8.331	33.3	8.333K	100.0K
	W01	95.6	95.6	44.0	44.0	75.7	75.7
	wo2	119.0	119.0	50.3	50.3	81.05	81.05
	wo3	148.3	148.3	56.6	56.(	86.71	86.71
	W04	185.2	185.2	94.3	94.3	160.8	160.8
	wo5	231.;	231.5	131.9	131.1	181	181
	α D1	1.20	3.12	1.00	0.10	0.6	0.08
	α D2	1.20	1.12	0.60	0.20	0.8	0.08
	α D3	1.20	1.12	1.00	0.10	0.6	0.08
	α D4	1.20	1.12	1.20	0.20	1.2	0.1
	α D5	1.20	1.12	1.20	0.30	1.2	0.1
	α N1	a.12	1.20	0.10	1.00	0.08	0.6
	α N2	0.12	I.20	0.20	0.60	0.08	0.8
	α N3	0.12	I.20	0.10	1.00	0.08	0.6
	d N 4	0.12	I.20	0.20	1.20	0.1	1.2
	α N5	0.12	I.20	0.30	1.20	0.1	1.2
	$\frac{\alpha N1}{\alpha D1}$	1/10	10	1/10	10	2/15	15/2
	$\frac{\alpha N2}{\alpha D2}$	1/10	10	1/3	3	1/10	10
	$\frac{\alpha N3}{\alpha D3}$	1/10	10	1/10	10	2/15	15/2
	$\frac{\alpha N4}{\alpha D4}$	1/10	10	1/6	6	1/12	12
	$\frac{\alpha N5}{\alpha D5}$	1/10	10	1/4	4	1/12	12
*	43R	10K	10K	10K	10K	8.333K	8.333K
	44R	10K	10K	10K	10K	8.333K	8.333K
**	53R	10K	10K	10K	10K	8.333K	8.333K
	54R	10K	10K	10K	10K	8.333K	8.333K

Figure 8, Sheet 2
Five Stage Cascade Band-Reject Filter

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